MACHINE LEARNING

PORTFOLIO RISK ANALYSIS AND INVESTMENT RECOMMENDATION SYSTEM USING PCA

DEPARTMENT OF CS&IT | KLH UNIVERSITY

Team and Project Guide

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Abstract

- Python + Streamlit app that analyzes Indian NSE stocks.
- Turns historical (or synthetic) prices into risk and return features.
- PCA reduces the features; K-means and hierarchical clustering group similar stocks.
- Logistic Regression and Random Forest predict Low / Medium / High relative risk.
- Shows example portfolios, a NIFTY 50 comparison and a backtest.
- Educational tool, not investment advice.

24 NSE stocks in the default universe

≥90% feature variance kept by PCA

3 portfolio weighting methods

Introduction

Data Overload

Markets create large amounts of price data every day.

Risk Matters

Investors need tools to measure risk and market behaviour.

ML Finds Patterns

Machine learning can reveal hidden patterns in financial data.

PCA Simplifies

PCA cuts many features to a few while keeping key information.

Problem Statement & Motivation

The Problem

- Each stock has many risk numbers (volatility, drawdown, skewness) that are hard to compare.
- It is hard to see which stocks behave alike.
- Picking stocks without grouping can weaken diversification.

Motivation

- Compress many metrics into a few PCA components.
- Group similar stocks to guide diversification.
- Give relative Low / Medium / High risk labels.
- Compare example portfolios with NIFTY 50.

Objectives

1. Collect NSE stock prices from Yahoo Finance or synthetic data
2. Calculate risk and return features for each stock
3. Reduce features with PCA, keeping at least 90% variance
4. Group stocks using K-means and hierarchical clustering
5. Classify relative risk as Low, Medium or High
6. Show example portfolios, compare with NIFTY 50 and backtest

Existing System and Limitations

Typical Existing Approaches

- Manual spreadsheets
- Single metrics such as volatility alone
- Stock-by-stock charts without grouping

Limitations

- Hard to handle many variables at once
- No clear groups of similar stocks
- Little guidance for diversification
- Results are hard to repeat or trace

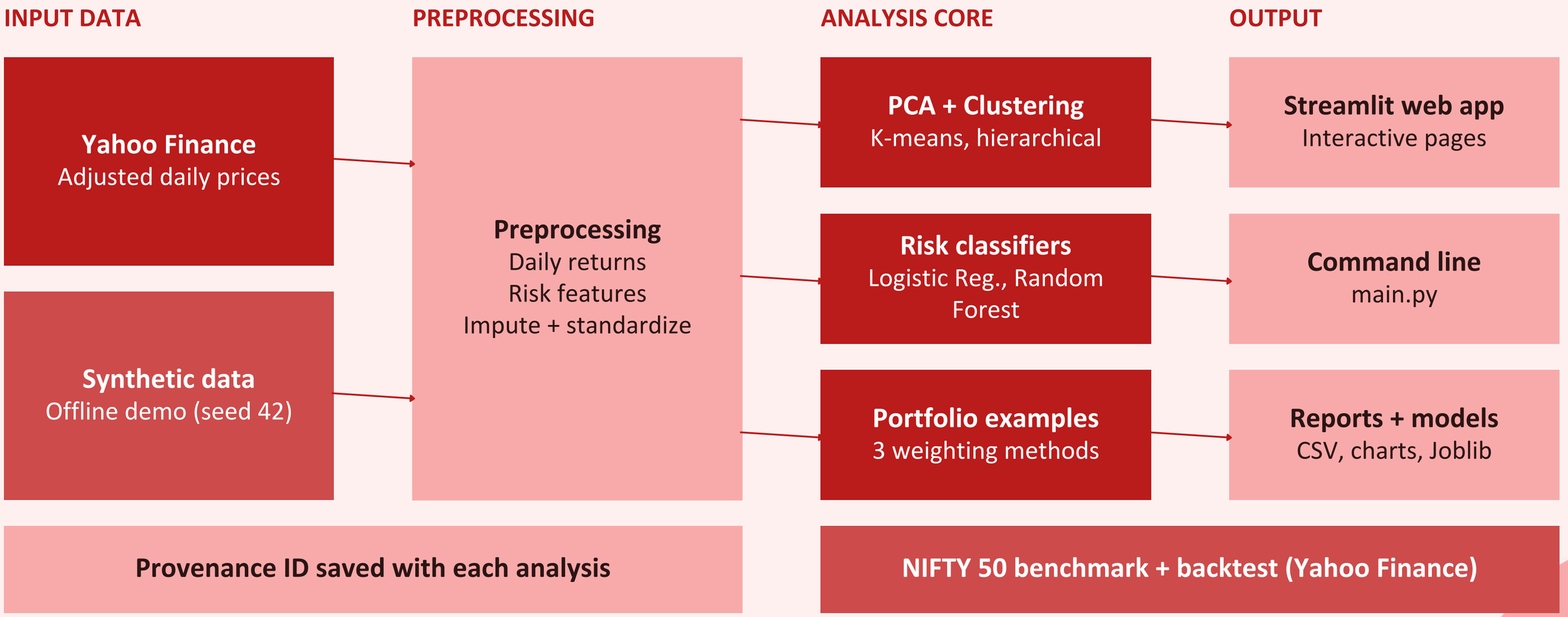
Proposed System

Python + Streamlit app

Works with Yahoo Finance data or offline synthetic data. Outputs are educational examples only.

- Risk and return features for each stock
- PCA with K-means / hierarchical clustering
- Low / Medium / High risk classification
- Equal, inverse-volatility, minimum-variance portfolios
- NIFTY 50 comparison and walk-forward backtest

System Architecture



Methodology and Workflow

1
Collect prices
Yahoo Finance or synthetic

2
Returns and features
Daily returns, 7 risk features

3
Impute + scale
Mean imputation, standardize

4
PCA
Keep at least 90% variance

5
Clustering
K-means and hierarchical

6
Risk classification
Terciles; LR and Random Forest

7
Portfolio examples
Equal, inv-vol, min-variance

8
Benchmark + visuals
NIFTY 50, backtest, charts

Technology Stack & Requirements

Requirements

- **Operating System:** Windows 10/11
- **Language:** Python 3.10+
- **IDE:** Visual Studio Code
- **Version Control:** Git & GitHub
- **Internet:** needed for Yahoo Finance mode

Pandas
Data tables

NumPy
Numerical maths

Scikit-learn
PCA, clustering, models

Matplotlib
Charts

yfinance
Price download

Joblib
Save fitted models

Streamlit
Web interface

System Modules & Implementation

Module	Role
`data_loader.py`	Yahoo Finance or synthetic prices
`preprocess.py`	Daily returns and risk features
`pca_pipeline.py`	PCA and feature loadings
`clustering.py`	K-means, hierarchical, validation scores
`classifier.py / evaluate.py`	Risk labels, LR and Random Forest, metrics
`portfolio_optimizer.py`	Equal, inverse-volatility, minimum-variance
`benchmark.py / backtesting.py`	NIFTY 50 metrics, walk-forward backtest
`pipeline.py / provenance.py`	Shared workflow and reproducibility metadata
`app.py / main.py`	Streamlit interface and command line

Dataset and Input Features

24	NSE stocks configured by default
2019–2026	Default date range (end date excluded)
Yahoo	Adjusted daily prices (needs internet)
Seed 42	Synthetic mode: offline demo data
NIFTY 50	^NSEI benchmark from Yahoo Finance

Input features per stock

- **Annualized return**
- **Annualized volatility**
- **Sharpe-like ratio**
- **Maximum drawdown**
- **Skewness**
- **Kurtosis**
- **5th-percentile daily return**

Algorithms and ML Models

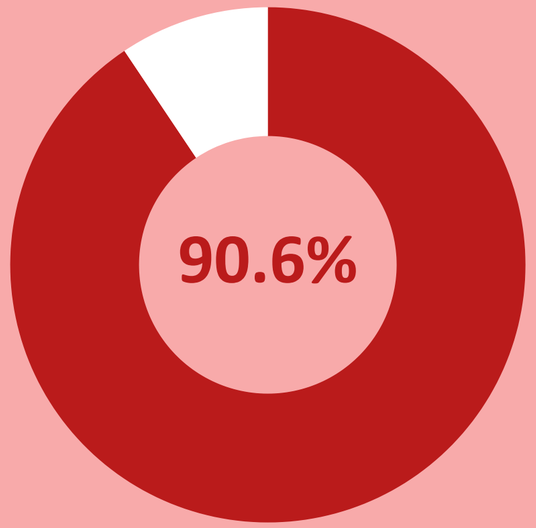
PCA	Reduces standardized features to a few components that keep at least 90% of variance.
K-means	Groups stocks into k clusters (k from 2 to 6) on PCA scores; k chosen by silhouette score.
Hierarchical	Agglomerative clustering, used to compare with the K-means groups.
Logistic Regression	Linear classifier that predicts the Low / Medium / High risk label.
Random Forest	Many decision trees vote on the label; also gives feature importance.

Unsupervised: PCA, clustering | Supervised: Logistic Regression, Random Forest

Results and Model Evaluation

EXAMPLE RESULT

PCA variance retained

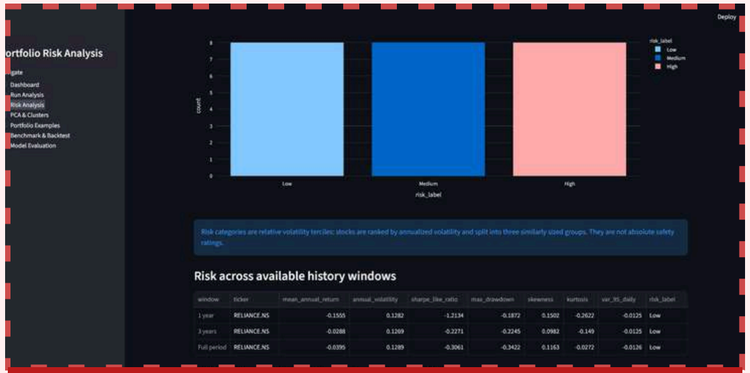


Previous synthetic run: 3 components. Example only.

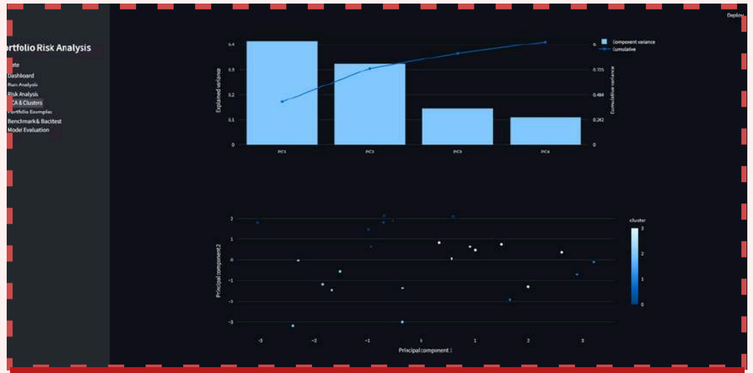
Model	Accuracy	Weighted F1
Logistic Regression	*[add from run]*	*[add from run]*
Random Forest	*[add from run]*	*[add from run]*

- Also reported: balanced accuracy, weighted precision and recall, confusion matrix.
- With about 24 stocks, scores are unstable and do not prove future prediction.

Application Screenshots / Demo



Dashboard



Risk Analysis



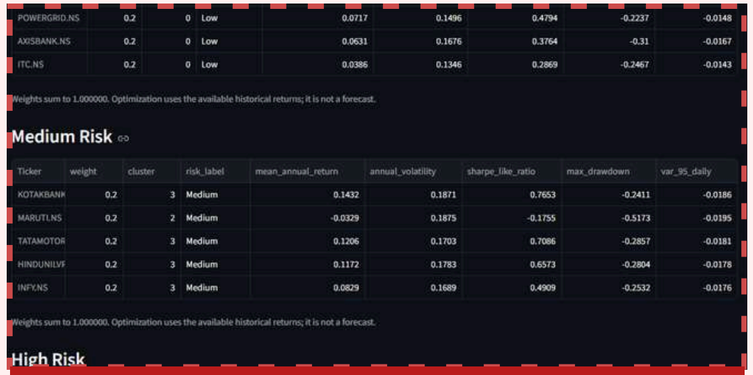
	PC2	PC3	PC4
	0.5642	0.1316	-0.0442
	0.3355	-0.5341	0.0165
	0.5552	0.1839	-0.033
	0.2632	0.5658	0.1274
	-0.2982	-0.0262	-0.455
	-0.1685	-0.0321	0.8774
	-0.2725	0.5845	-0.0601

PCA & Clusters

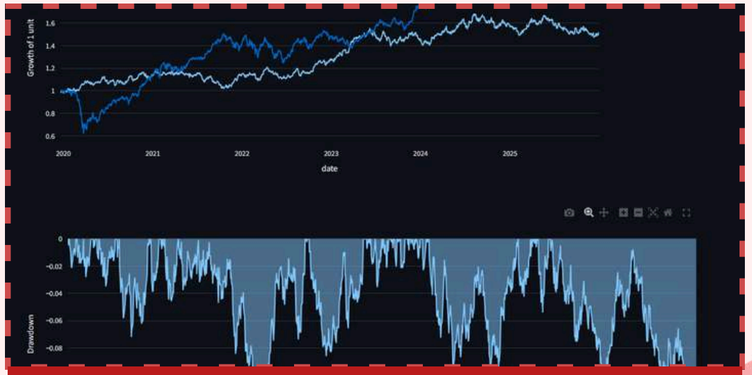


	PC1	PC2
0	-3.0562	
0	-0.9805	
3	0.3384	
3	0.5666	
1	2.9032	
1	3.2151	
0	-0.6945	
0	0.5993	
3	0.9123	
3	2.6192	

Portfolio Examples



Benchmark & Backtest



Model Evaluation

Placeholders: no screenshots included. Add real ones from your own run.

Advantages and Applications

Advantages

- PCA simplifies many risk metrics
- Groups similar stocks to aid diversification
- Interactive Streamlit dashboard
- Reproducible: provenance ID and saved artifacts
- Works offline with synthetic data

Applications

- Learning machine learning in finance
- Academic and course projects
- Exploring NSE stock risk profiles
- Teaching diversification ideas

Limitations

- Historical risk and return do not predict future results
- Fixed 24-stock list: selection and survivorship bias
- Risk groups are relative (volatility terciles), not absolute ratings
- Classifier scores are unstable with only about 24 stocks
- Sector, liquidity, taxes and real trading costs are not fully modeled

Portfolio outputs are educational examples, not guaranteed investment advice.

Future Scope

PROPOSED ENHANCEMENTS: NOT YET IMPLEMENTED

More models

- Gradient boosting (XGBoost, LightGBM)
- Ridge and Lasso regression
- DBSCAN and t-SNE views

Better evaluation

- ROC-AUC and nested cross-validation
- Hyperparameter search
- Permutation importance or SHAP

Data and deployment

- Point-in-time stock universe, sector and liquidity limits
- FastAPI prediction service
- Drift monitoring, scheduled retraining

Conclusion

- Built a complete ML workflow for NSE stock risk analysis in Python and Streamlit
- PCA simplifies the risk features; clustering groups similar stocks
- Logistic Regression and Random Forest label relative risk
- Includes example portfolios, NIFTY 50 comparison and a chronological backtest
- An educational tool: results are examples, not investment advice

Thank You...

Any questions?